Villanova University | **Departments of Physics** | **Fall - 2025**
Physics 1101 | | **All Sections**

In person Laboratory | **Last update: August 7, 2025**

Announcements: all communications, submission of reports, etc. in regard to the course will be conducted through the Blackboard's (Bb) site of the course. To go to the Bb site:

1. go to Villanova site,
2. click on the ELEARNING link,
3. login using your email ID and PW,
4. and from the list of your courses pick PHY1101.

Bb site- PHY1101

All lab sessions are in person

- Location: Mendel Hall 257
- Instructor: See the table below
- Office: See the below
- Email: See the table below

Fall session: the fall session of phy1101 labs starts on the week of Monday, August 25. No labs on the week of September 1.

Lab Manual: is available as a set of separate files to the students registered in the course at the Blackboard(Bb) Content Homepage under **Experiments**.

PASCO Capstone Software: We are using PASCO's Capstone(CS) as the software to import and analyze data. CS will be used extensively in accessing the data, its presentation and analysis. The students are required to download the software on their personal laptops prior to the first laboratory session (week of August 25, 2025). To download directly from the PASCO site visit:

PASCO'S CAPSTONE DOWNLOAD
The key for registering and running the licensed software can be found in a text file on the course's Bb site.

Lab Attendance: Active participation in all live sessions is required. Attendance will be taken and recorded at the beginning of each session.

Course Content: Selected experiments in mechanics will be performed, with emphasis on the acquisition and analysis of the experimental data. Students are expected to complete all the necessary work in the laboratory setting to write a comprehensive lab report. This sequence of laboratory experiments is designed to complement the co-requisite lecture course Physics 1100.

Goals: In addition to the traditional goals of observation and analysis of physical data relevant to some classical experiments in physics, students should, upon successful completion of this lab, be proficient in using the computer as an integral part of experimental apparatus. They should also develop sufficient skills in using data collection and analysis software, specifically PASCO's Capstone. While Capstone may not be used in future experimental work, it should serve as a useful prototype for other applications. Equally important, students should develop the skill to maintain a detailed lab workbook, enabling them to write a clear and coherent scientific reports to present their work.

Reports: individual reports should be submitted electronically, following the specific requirements provided by the instructor. Reports must be written independently, and each student must submit their report separately.

Grades: for the lab will be determined based on:
(1) Active participation in all sessions,
(2) The submission of reports,
(3) Compliance with all the other requirements outlined by your instructor

We will go over the above details many times during the term.

The Office of Access and Disabilities Services: students with disabilities who require reasonable academic accommodations should schedule an appointment to discuss specifics with this office. It is the policy of Villanova to make reasonable academic accommodations for qualified individuals with disabilities. You must present verification and register with the Learning Support Office by contacting **610-519-5176** or at **VULSS** for physical access or temporary disabling conditions, please contact the Office of Access and Disability Services at **610-519-4095** or email

Falvey Test Center.

Registration is needed in order to receive accommodations.

Academic Integrity: all students are expected to uphold Villanova's Academic Integrity Policy and Code. Any incident of academic dishonesty will be reported to the Dean of the College of Liberal Arts and Sciences for disciplinary action. For the College's statement on Academic Integrity, you should consult the Enchiridion. You may view the university's Academic Integrity Policy and Code, as well as other useful information related to writing papers, at the Academic Integrity Gateway web site:

OFFICE OF ACADEMIC INTEGRITY

Grading scale:

Grade	Range of the Grade
A	≥ 92.5
A-	$90-92$
B+	$87.5-89.5$
B	$\geq 82.5-87.0$
B-	$80-82$
C+	$77.5-79.5$

Grade	Range of the Grade
C	$\geq 72.5-77$
C-	$70-72$
D+	$67.5-69.5$
D	$\geq 62.5-67$
D-	$60-62$
F	≤ 59.5

Sections and Instructors

Section	Time	Instructor	Office
08.	M. 8:30 – 11:20	**Javad Siah**	M263C
01.	M. 1:55 – 4:45	**Javad Siah**	M263C
02.	T. 8:30 – 11:20	**Javad Siah**	M263C
03.	T. 1:00 – 3:50	**Morgan Besson**	M367B
100.	T. 6:15 – 9:05	**Vianney Gimenez Pinto**	M365C
07.	W. 8:30 – 11:20	**Vianney Gimenez Pinto**	M365C
04.	W. 1:55 – 4:45	**Morgan Besson**	M367B
101.	W. 6:15 – 9:05	**Deepika Khilnaney**	M284B
05.	R. 8:30 – 11:20	**Maia Magrakvelidez**	M344
06.	R. 1:00 – 3:50	**Georgia PapaefthymiouDavis**	M367C
09.	F. 10:40 – 01:30	**Scott Dietrich**	M263A

Laboratory Schedule: Experiments

Experiment	Title	Date - week of
1.	Introduction to the computer and analysis techniques: parts I. II. and III.	August 25
2.	The simple pendulum	September 8
3.	The simple pendulum, measurement of “g” and error analysis	September 15
4.	Kinematics - Part I.	September 22
5.	Kinematics - Part II.	September 29
6.	Free Fall	October 6
7.	Kinetic and Static Friction	October 20
8.	Newtonian Dynamics	October 27
9.	Impulse, momentum and conservation of linear momentum	November 3
10.	Hooke’s law, elasticity and periodic motion	November 10
11.	The simple harmonic motion	November 17
12.	Make-up Labs. One experiment at discretion of the instructor	December 1